# Theophilus C. Onyejiaku

## Machine Learning Engineer | Computer Vision · RAG & LLM Apps · Model Deployment

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Machine Learning Engineer who builds, fine-tunes and ships models end to end. Delivered a YOLOv11n tomato quality detector running on Android at 189 ms inference (95.4% mAP50), a CNN flower-recognition web app published at an IEEE conference, and a retrieval-augmented PDF chat application. Backed by full-stack skills in Flask, MySQL and Flutter, plus data analytics and technical writing experience.

## CORE SKILLS

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|---|---|
| **ML & Deep Learning** | TensorFlow, Keras, YOLO (v11), CNNs, Scikit-Learn, OpenCV, TensorFlow Lite, Hyperparameter Tuning, Data Augmentation, Model Evaluation & Monitoring |
| **GenAI & LLMs** | Retrieval-Augmented Generation (RAG), OpenAI API, DALL-E, LLMOps, Prompt-based Applications |
| **Engineering** | Python, Flask, REST APIs, JWT Auth, MySQL, SQLAlchemy, Flutter / Dart, Streamlit, Git & GitHub, PyInstaller |
| **Data** | Pandas, NumPy, Matplotlib, SciPy, SQL, Power BI, Tableau, MATLAB, R, C++ |
| **Platforms** | Jupyter Notebook, Google Colab, Kaggle Notebook, TensorBoard, PyCharm, VS Code |

## MACHINE LEARNING EXPERIENCE

**Graduate Research Assistant** **Real-Time Tomato Quality Detection (YOLOv11n, Android)** — Jan 2025 – Present

- Developed a real-time Android tomato quality detector using YOLOv11n to classify tomatoes as healthy, defective or unripe.
- Curated and annotated 2,400 images; fine-tuned for 50 epochs to reach mAP50 of 95.4% and mAP50–95 of 86.3% (precision: 95.4% healthy, 92.5% defective, 98.4% unripe).
- Optimised the model with TensorFlow Lite and deployed it on Android with 189 ms average inference time. Manuscript under review.

**Graduate Research Assistant** **Flower Recognition & Status Monitoring Web App** — Aug 2023 – Sep 2024

- Built and trained a CNN (TensorFlow, Keras) to classify multiple flower species, tuning hyperparameters for precision and recall.
- Collected, preprocessed and augmented image data to generalise across lighting and background conditions.
- Deployed the model in a Streamlit dashboard showing predictions, confidence scores and growth-status metrics, with continuous monitoring and performance logging.
- Co-authored the resulting paper, published at IEEE ICTIIA 2024; live demo: Streamlit app.

## SELECTED PROJECTS

**RAG PDF Chat** **Retrieval-Augmented Generation** — Python

- Built an application that lets users chat with their PDF documents, with answers grounded in the uploaded content through retrieval-augmented generation.

**Morning Star CBT** **Offline Computer-Based Testing System** — Flask · MySQL · Flutter

- Built an offline, LAN-based exam platform for a school supporting 40 students across 14 subjects, with a Flask/MySQL backend, JWT authentication, SQLAlchemy models and a Flutter web client.
- Packaged the backend as a Windows executable with PyInstaller; added an exam-retake workflow with a permanent audit log and class-filtered CSV result exports.

**AI Video Generator** **Freemium Web App** — Flask · FFmpeg

- Designed an app that turns a topic into a finished video: AI-written script, text-to-speech narration and stock footage, assembled automatically with FFmpeg.

**Data Professionals Salary Prediction Machine Learning Web App** — Python

- Performed EDA and built a prediction model on data-professional salaries, deployed as a web application (EDA repo).

**Kaggle Notebooks** **Model Tuning & Evaluation** — Python

- Published notebooks: Model Tuning, a Different Approach and Classification Reports of Prediction Models.

## ADDITIONAL EXPERIENCE

**Artificial Intelligence Instructor (Part-time)** Franklin Comprehensive — Sep 2024 – Jul 2025

- Taught data preparation, machine learning models and computer vision applications using Python, Streamlit, TensorFlow and OpenCV.

**Data Analyst** Medbury Medical Services; RKY Consulting Services — Apr 2022 – Present

- Analysed datasets of 10M+ rows with SQL and Python, automated dashboards and trained 400+ learners in data analytics tools.

**Technical Content Writer (Freelance)** Comet, InfluxData, Tricentis, Autify — Oct 2021 – Aug 2026

- Authored 660+ technical articles on machine learning, AI and developer tools; published work ranks on Google's first page.

## EDUCATION

**B.Eng. Agricultural & Bioresources Engineering** — 2015 – 2022

*Federal University of Technology Owerri, Nigeria | CGPA 4.20 / 5.0 | Best Graduating Student*

## CERTIFICATIONS

- LLMOps: Building Real-World Applications with Large Language Models – Udacity
- Computer Vision, Intro to Deep Learning, Intermediate Machine Learning, Feature Engineering, Pandas – Kaggle

## PUBLICATION

T. C. Onyejiaku, C. I. Nwakanma and B. C. Ekeoma, "Deep Learning-Based Web Application for Flower Recognition and Status Monitoring," 2024 2nd International Conference on Technology Innovation and Its Applications (ICTIIA), Medan, Indonesia. IEEE. doi:10.1109/ICTIIA61827.2024.10761803